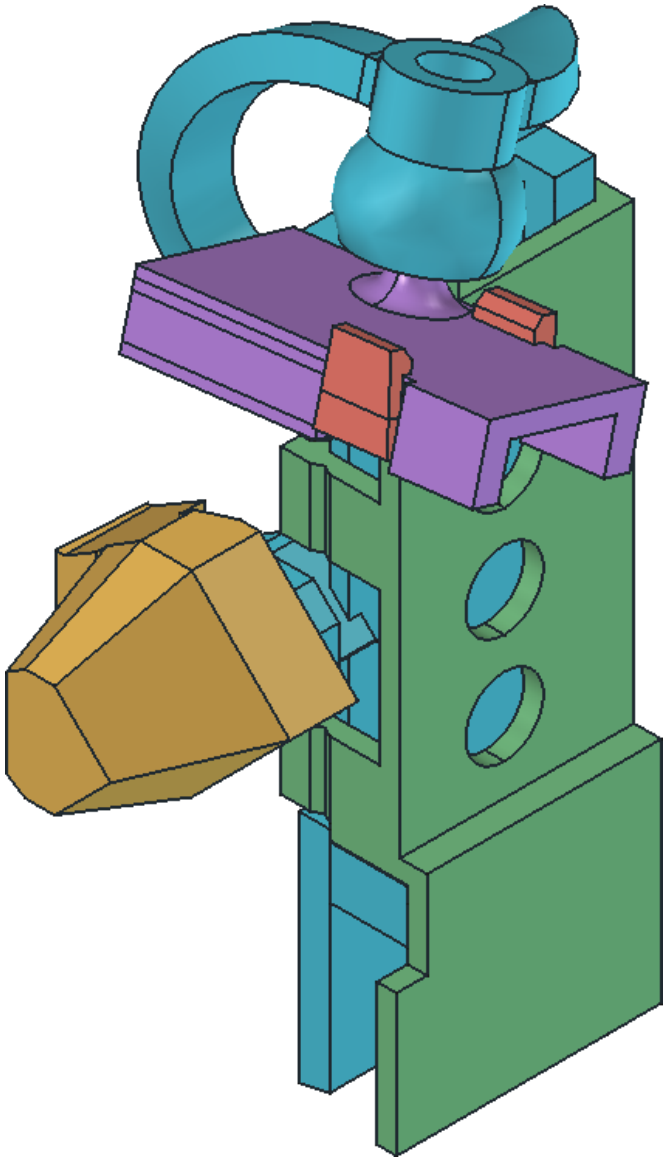
A clear starting point

Five named parts. One assembled coordinate frame. Existing geometry preserved.



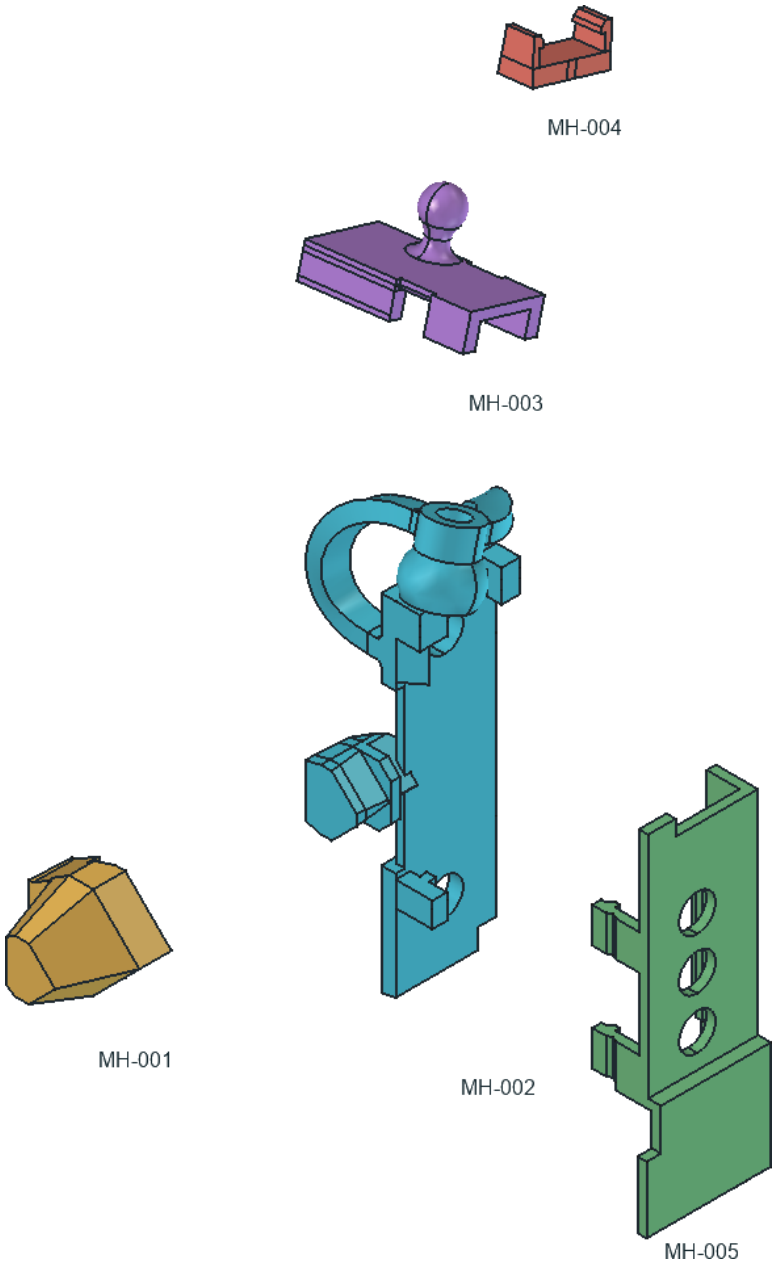
Open MouseHats_R02.step for neutral CAD exchange (AP242, millimeters).
Open MouseHats_R02.FCStd for the same clean, colored assembly in FreeCAD.
Both are geometric snapshots. Editable feature documents live under source/.
Use print/ for the four intended STL inputs and the printing notes.

What changed in R02?
Only the baseplate lower exterior: tapered to a smaller contact footprint.
R01 carrier, holder, clip and cap geometry is preserved. Earlier work remains
in the playground; this folder is the clearly identified sharing checkpoint.

Status: physical fit, strength and joint release still need a print test.

Parts and relationships

Exploded reference only - offsets are for illustration, not assembly dimensions.



MH-001 Baseplate (gold): receives the carrier stem; tapered exterior.

MH-002 Carrier (cyan): PCB tray + stem + both arms + socket; one solid.

MH-003 Camera holder (purple): sensor cradle, neck and ball; one solid.

MH-004 Retaining clip (red): wraps the thickened holder walls.

MH-005 PCB cap (green): retains/covers PCB and connector region.

Interfaces: stem/baseplate; ball/socket; holder/clip; tray/cap; PCB/retainers.

Electronics, ribbon, screws, inserts and skull are omitted, not modeled as solids.

Print carrier + holder together in the supplied pose; print accessories separately.

Controls and design intent

Edit a working copy of source/. Keep this published checkpoint unchanged.

Carrier / LongerPCBRetainers

Two arm-end retainers extend 1.5 mm along the board edge; lower stop 0.7 mm.
Slot gap and PCB seating position stay fixed. The lower extension preserves cap fit.

Holder / ThickerHolderWalls + ExtendedClip

Walls grow outward by 0.5 mm per side; inner sensor geometry stays fixed.
Clip span grows 1.0 mm with intact retaining arms. These controls are coupled.

Baseplate / TaperedBaseplate

BottomScale = 0.57004. Taper from Y = 1.5 mm to Y = -5 mm.
Contact area: 67.483 to 21.928 mm2. Mount region at/above Y = 1.5 is preserved.

Carrier / JunctionFillets + BlendedCarrier

Tray/arm roots R0.5, socket/arm junctions R0.3, stem/tray blend R3 mm.
These changes address observed fragility; strength has not been load-tested.

How to explore

1. Run source/Open-editable.FCMacro in FreeCAD 1.1.4 to load proxy modules and the editable carrier, holder/clip, baseplate and cap documents.
2. Change one named control in a working copy, recompute, inspect mating parts.
3. Use the included rebuild_r02.py with FreeCAD Python to regenerate R02 edits from the included parametric checkpoint. See source/README.md.
4. Revalidate interfaces before creating the next revision.

Source provenance: reconstruction from supplied 58 STEP; two-arm paths and holder pose informed by V5 mesh; baseplate taper measured from tapered 53 mesh.
The raw historical meshes/Fusion archives are not duplicated in this package.

Evidence and open questions

A release checkpoint is traceable; it is not a claim that every fit is validated.

Verified in this package

Five named, closed solids import from STEP; assembly labels and colors survive.
Volume error < 0.0001 mm3; centroid/bounds error < 0.00001 mm against native CAD.
The deeper STEP Boolean-readiness check still reports InvalidCurveOnSurface warnings on four parts. Basic solid validity passes. See step-roundtrip.json.
Treat STEP as a qualified exchange candidate; native CAD is included as fallback.
A second CAD application check has not been performed.

Existing design checks

Carrier/holder have zero overlap and about 0.121 mm minimum modeled separation.
A tiny clip/holder contact (~0.0045 mm3 at the recorded Boolean tolerance) remains.
The new baseplate preserves the upper mount region and has no carrier overlap.
These are geometric checks, not printer tolerances or a full motion analysis.

First print questions

- Does the print-in-place ball release without breaking the neck or holder?
- Do the clip, PCB and cap install securely without excessive force?
- Does the ribbon route without binding through the desired camera motion?
- Do support removal and handling leave the arm roots and stem intact?
- Is the smaller baseplate footprint suitable for the intended mounting surface?

Not included in this revision

Larger ball/neck/socket, additional weight-relief holes, and PCB/cap vent changes.
No slicer profile, supports, printer job, measured fit or strength certification.

Collaboration: identify the part number, parameter, screenshot/view and revision when proposing a change. Explain the failure or intended benefit, and record which interfaces must remain fixed. See CHANGELOG.md and VALIDATION.md.